Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/07/2024
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into The Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/10/2017	**End Date:**	06/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2024 per F. Hernandez/ARC (Ed., 7/5/23) NOTE: End date changed to 6/30/2023 per NSSC (Ed., 5/16/22) NOTE: End date changed to 6/30/2022 per F. Hernandez/ARC (Ed., 8/19/21) NOTE: End date changed to 7/31/2021 per F. Hernandez/ARC (Ed., 6/4/2020) NOTE: Extended to 7/31/2020 per NSSC information (Ed., 5/15/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schug, Malcolm (UNC Greensboro)		
Grant/Contract No.:	80NSSC17K0546		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	[ED. NOTE 12/11/2017: Continuation of "Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants--NNX12AO65G," per PI move from University of Mississippi to University of North Carolina at Greensboro.] The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the International Space Station (ISS) with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis. In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers will be launched via Space-X, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying the effects of gravity and light on basic cellular mechanisms in plants, this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	The future of space exploration will be contingent upon the use of plants in bioregenerative life support systems. Unfortunately, the microgravity of space can cause stress in plants, which can reduce growth. The Random Positioning Machine (RPM) is a device designed to provide an analogue for the effects of microgravity on Earth by rotating in three dimensions. In this study, we compare the results from experiments conducted on the International Space Station with those conducted using the RPM (in the 3D clinostat mode) on the ground. Seedlings of Arabidopsis thaliana wildtype and phytochrome mutants were grown in true microgravity and in the omnidirectional gravity on a rotating RPM on the ground. We found that the RPM treatment caused less stress in the seedlings than did true microgravity. We also report that phytochromes A and B play roles in phototropic responses to unilateral light and that these roles differ in the two gravitational environments. Finally, we conclude that while root phototropism in unilateral red and blue differs significantly between the microgravity and omnidirectional stimuli, the RPM can serve as a reasonable analogue of microgravity conditions for assessment of shoot phototropism.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Jenkins B, Hughes AM, Bauer L, Barker R, Kiss JZ. "Quantitative phenotypic analysis of Arabidopsis thaliana grown in microgravity using SOAP, an applied artificial intelligence." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Hughes AM, Vandenbrink JP, Kiss JZ. "Efficacy of the Random Positioning Machine as a terrestrial analogue to microgravity in studies of seedling phototropism." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Articles in Peer-reviewed Journals	Hughes AM, Vandenbrink JP, Kiss JZ. "Efficacy of the Random Positioning Machine as a terrestrial analogue to microgravity in studies of seedling phototropism." Microgravity Science and Technology. 2023 Aug 14;35(4):43. https://doi.org/10.1007/s12217-023-10066-9 , Aug-2023
Significant Media Coverage	Varmazis M (Kiss JZ interview). "Plants in space; plants on the ISS; plants on the Moon." Podcast. T-Minus Space Daily, September 5, 2023. https://space.n2k.com/podcasts/t-minus/108 , Sep-2023
Significant Media Coverage	Livingston DM (Kiss JZ interview). "Space settlement and plant space biology." The Space Show podcast. https://www.thespaceshow.com/guest/dr.-john-z.-kiss-0 , Aug-2023